

Divanshu Sharma

AI Engineer

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SKILLS

Languages & Libraries:	Python, SQL, C++; PyTorch, TensorFlow, Scikit-learn, Keras, NumPy, Pandas
Data Engineering & Automation:	Data Mining, Workflow Automation, Data Quality & Integrity, Process Scripting; Airflow, FastAPI, MLflow; Web Scraping (BeautifulSoup-compatible pipelines)
LLMs & Agentic AI:	LangChain, Hugging Face Transformers, RAG-based Agents, Agentic AI Architectures, Prompt Engineering, LLM Evaluation Metrics, GPT & Claude Models, AutoGen/ReAct Patterns, DeBERTa, RoBERTa, BioMistral7B, Whisper, SFT
Vector Stores & Data:	Vectorization, Vector Stores (Qdrant, Chroma), PostgreSQL, MySQL, MongoDB; EDA, Statistical Analysis, Regression Analysis, Data Mining, ETL Pipelines
MLOps & Cloud:	Docker, CI/CD (GitHub Actions), MLflow, Airflow, FastAPI, AWS (SageMaker, EC2), GCP, Azure
Developer & Visualization Tools:	Git, Jupyter, Postman, Colab; React, Next.js, Server-Sent Events; Data Visualization, Analytical Storytelling

EXPERIENCE

Uniiq.ai | Founding Engineer | May 2026 - Present

- Architected and shipped a production-grade Generative AI application end-to-end as the sole engineer, developing a conversational LLM onboarding agent that streamlines profile creation, university matching, and application tracking for 250+ active users.

Yale University | ML Researcher | Oct 2025 - Mar 2026

- Engineered a secure, open-source deep learning library from first principles (Python/NumPy, no external autograd), implementing a compiler-centric MPC architecture delivering verifiable cryptographic security for privacy-preserving collaborative AI model training.
- Designed and trained a custom CNN with strided convolutions, achieving 88.08% test accuracy on the ChestMNIST benchmark and performing failure-mode and generalization analyses to assess model robustness.

WorldQuant BRAIN | Research Consultant | Nov 2022 - July 2025

- Built and automated **15+** high-frequency alpha signal generation and backtesting workflows using Python and WorldQuant Simulation, replacing manual analysis and achieving an average Sharpe Ratio of 1.8.

PROJECTS & OPEN-SOURCE

Graph Data Explorer AI | [GitHub](#) | [Link](#)

Technologies: LangChain, LangGraph, FastAPI, Qdrant, Knowledge Graphs, Server-Sent Events, React, NetworkX

- Built an end-to-end data engineering pipeline** transforming **21,393 transactional records across 19 relational tables** into a 765-node / 877-edge NetworkX knowledge graph; implemented a **two-stage LLM orchestration pipeline** (NL → Python → Sandboxed Execution → NL) with query guardrails — directly mirroring Visa's need for automated orchestration of data extraction logic.
- Delivered a production **FastAPI + React full-stack application** with real-time streaming via Server-Sent Events; maintained **code quality through review cycles**, unit testing, and component-level design following SDLC standards.

End-to-End Indic ASR Optimization & Consensus Evaluation | [GitHub](#)

Technologies: Whisper, SpecAugment, Beam Search, Devanagari NLP, MLflow, Docker, FastAPI

- Achieved **>48%** absolute WER reduction on the FLEURS benchmark by fine-tuning Whisper-small with SpecAugment, Label Smoothing, and Cosine LR scheduling evaluated via a 5-beam Beam Search pipeline.
- Reduced transcription errors across **177,000** unique terms by building a Devanagari normalization engine and a phonetic reverse-transliteration layer, fusing 5 ASR models via Word-level Confusion Networks with majority-voting.
- Fused 5 ASR models via Word-level Confusion Networks; ML model building, pipeline orchestration, and rigorous quantitative evaluation demonstrating end-to-end automated data processing at scale.

RenAIssance OCR - Research-Grade Document Intelligence Pipeline | [GitHub](#)

Technologies: CRAFT, SE-ResNet-BiLSTM CRNN, RoBERTa, LLM Post-Processing, Docker, FastAPI, Azure

- Reduced CER from 3.44% to 0.00% on a 183-sample validation set by chaining CRAFT text detection, an SE-ResNet-BiLSTM CRNN recognizer, and an LLM-based RoBERTa post-processor
- Scaled the pipeline to 12,792 training samples of degraded 16th–17th century Spanish texts using advanced data augmentation and a 7-step preprocessing workflow, lifting full-corpus Word Accuracy from 97.96% to 98.17%.

EDUCATION

University of Mumbai | Bachelor of Engineering, Computer Science | June 2025 | CGPA: 8/10.0

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Distributed Systems, System Design

ACHIEVEMENTS & CERTIFICATIONS

- GOLD LEVEL (Top 1 %), STAGE 1 Qualifier, WorldQuant BRAIN Global Alphathon 2022: Achieved a top-tier ranking among thousands of global participants by designing and submitting high-performing quantitative trading models.
- Earned Kaggle 2× Expert status; solved 3,000+ algorithmic problems across LeetCode (Knight), CodeChef (Max Rating 1662), and HackerRank.
- Completed ML Specialization, Deep Learning Specialization, TensorFlow Developer Certificate, and GANs Specialization (DeepLearning.AI, 2023–2024).